

Don Santos

don16santos@gmail.com • +27 67 921 5702 • Gauteng, South Africa • [LinkedIn](#) • [GitHub](#)

PROFESSIONAL SUMMARY

Systems Architect and Full-Stack Engineer specializing in designing fault-tolerant AI orchestration pipelines, distributed multi-agent swarms, and high-performance React architectures. Expert in resolving the compounding error rates of single-instance large language models through system-level redundancy, consensus-based voting protocols, and recursive task decomposition. Partnering with startups and founders to build production-grade, deterministic AI platforms and user interfaces that maximize throughput and visual excellence.

PROFESSIONAL EXPERIENCE

Full-Stack Architect & Creator | DocketDive

- Architected and built DocketDive, an industrial-grade, AI-native legal research assistant designed to replace unstable, single-agent search queries with a 5-worker consensus Swarm Engine.
- Designed and implemented Maximal Agentic Decomposition protocols, splitting complex document parsing and citation verification workflows into granular, low-token decisions to mathematically suppress model hallucinations.
- Created a Gold Standard PDF rendering pipeline leveraging headless Puppeteer and custom print stylesheets, ensuring pixel-perfect, ATS-optimized, Harvard-style document exports.
- Engineered a multi-threaded batch scraping pipeline that tails court rolls, dynamically customizes research digests, and updates vector indexes with redundant worker verification.
- Developed a real-time Quality Intelligence service providing semantic scoring, metric-driven citation suggestions, and legal research feedback using local LLM models.

Student Assistant | University of Johannesburg

- Managing Student Marks: Responsible for accurately collecting, organizing, and entering student marks into a database, ensuring data integrity.
- Handling Large Workloads: Effectively managing and prioritizing a high volume of student marks, assignments, and projects, demonstrating strong organizational skills and attention to detail to ensure timely completion.
- Accessing and Reviewing Advanced Accounts: Utilizing advanced account privileges to access confidential student information, reviewing records, and maintaining strict confidentiality and data protection protocols.
- Project Management: Collaborating with colleagues to coordinate and execute projects, including gathering necessary information, ensuring project milestones are met, and delivering completed work within specified timelines.

EDUCATION

Paralegal Studies | University of Johannesburg (UJ)

Full-Stack Certificate in Web Development | HyperionDev

Ongoing Research in Distributed Systems & LLM Error Correction | Independent Engineering Research

SKILLS

AI & Systems Design: Swarm Intelligence, Ensemble Voting, Maximal Decomposition, RAG Architectures, Vector Databases, LangChain, Ollama / Groq

Frontend Engineering: React, Next.js 14, TypeScript, Tailwind CSS, Framer Motion, WebGL Shaders

Backend & Tooling: Node.js / Express, REST & WebSockets, SQLite / Postgres, Puppeteer Rendering, Docker, Git / CI-CD

Soft Skills: Written & Verbal Communication, Presentation, Team Collaboration, Work Under Pressure, Adaptability & Systems Learning

PROJECTS

DocketDive — AI Legal Research Assistant

[View Project](#)

South Africa's first AI-native legal research assistant. Created a multi-stage Retrieval-Augmented Generation (RAG) system that allows legal professionals to query, analyze, and cite case law with zero hallucinations.

StudyPodLM — Multi-Modal Learning Workspace

[View Project](#)

An interactive AI education platform converting textbooks, lecture notes, and video assets into structured study guides, complete with custom text-to-speech rendering and self-assessment engines.

Creative Artworks & Design Lab

[View Project](#)

A premium gallery showcasing custom photo manipulations, digital editing, and visual assets. Integrated with interactive layouts designed using Figma and Framer, demonstrating a strong foundation in visual storytelling, UI/UX design, and creative tools like Adobe Photoshop and Canva.